

SIHAN JIANG

New Haven, CT | (203) 640-6611 | sihan.jiang@yale.edu | www.linkedin.com/in/sihan-jiang-b7395a23b/

EDUCATION

Yale University, New Haven, CT, U.S.

08/2025-Present

M.S., Biostatistics (Data Science pathway)

- **GPA:** 3.925/4.0
- **Relevant Coursework:** Large Language Models, Intermediate Machine Learning, BiomedicalDataSci: Mining&Mdlng

Duke Kunshan University (DKU), Kunshan, Jiangsu, China

08/2021-05/2025

B.S., Computer Science (DKU-Duke Dual Degree)

- **GPA:** 3.811/4.0
- **Relevant Coursework:** Principles of Machine Learning, Design & Analysis of Algorithms, Cell and Molecular Biology
- **Awards & Honors:** Latin Honors - Cum Laude; Dean's List-Fall 2021, Spring 2022, Spring 2024

PUBLICATIONS & CONFERENCE PRESENTATIONS

- Liu, T., **Jiang, S.**, Zhang, F., Sun, K., Head-Gordon, T., & Zhao, H. (*Nature Communications* in Submission). DrugPlayGround: Benchmarking Large Language Models and Embeddings for Drug Discovery. (*Co-First Author*)
- **Jiang, S.**, & Huang, M. C. (2025). Diabetes Prevention Chatbots to Help Dietary Self-management for the Chinese Population. *Annual International Conference of the IEEE Engineering in Medicine and Biology Society. IEEE Engineering in Medicine and Biology Society. Annual International Conference, 2025*, 1–7. <https://doi.org/10.1109/EMBC58623.2025.11254311> (*First Author*)
- **Jiang, S.**, Shen J., & Chen M. (2024). Factors Included in AI Algorithms Assisting Precision Medicine in Cancer Treatment: A Systematic Review. Poster Presentation. *Consortium of Universities for Global Health (CUGH) Conference, 2024*. (*Co-First Author, Presenter*)

RESEARCH EXPERIENCE

DrugPlayGround: Benchmarking Large Language Models and Embeddings for Drug Discovery 05/2025-01/2026

LLM Research Assistant (Supervisor: Hongyu Zhao)

New Haven, CT, U.S.

- **Background:** Explored LLMs and embeddings' performance in cancer drug-centered analysis, e.g. function inference.
- **Data Preparation & Prompt Engineering:** Extracted and curated 4 datasets (1k+ cancer drug entries each) from PubChem and Hugging Face; Designed Normal, CoT, and Meta prompts for comparative analysis of LLMs.
- **LLM & Embedding Benchmark:** Developed Python pipelines on HPC clusters with Slurm and Linux for batch-generating 100k+ drug descriptions from 5 LLM APIs (GPT, Gemini, Claude, DeepSeek, Mistral) via parameter tuning, and embeddings with 5 Embedding Models (Gemma-300m, Gemini, OpenAI, Mistral, Qwen3).
- **Statistical Evaluation:** Implemented an evaluation framework calculating BERT, ROUGE, BLEU scores and cosine similarity to measure the quality of the output and their similarity against ground-truth data.
- **Data Visualization & Analysis:** Conducted statistical analysis (mean, SD, variance) to evaluate temperature sensitivity and model accuracy; Synthesized advanced visualizations (line plots, box plots, bubble heatmaps, etc.) for downstream prediction tasks, such as drug-protein interaction.
- **Task-specific Comparison:** Compared LLM-based embeddings' performance with domain-specific baselines (e.g. Uni-Mol, ChemCPA with RDKit) for drug-target interaction, drug synergy, and chemical perturbation prediction.

Diabetes Prevention Chatbots to Help Dietary Self-Management for the Chinese Population

06/2023-05/2025

Research Investigator (Supervisor: Ming-Chun Huang)

Kunshan, Jiangsu, China

- **Background:** Investigated the use of LLMs for culturally tailored digital health interventions, focusing on assisting diabetes prevention in Chinese populations via lifestyle self-management.
- **Data Preparation:** Curated a China-focused diabetes prevention guideline by synthesizing 5+ academic papers (emphasizing diet/lifestyle), and a specialized conversational JavaScript dataset with 80+ data based on it.
- **LLM Fine-tuning:** Fine-tuned GPT-3.5-turbo and GPT4o through OpenAI API using the prepared dataset to improve their performance on offering culturally contextualized, evidence-based prevention advice.
- **Model Evaluation:** Conducted quantitative and qualitative, scenario-based evaluations to assess response accuracy, guideline adherence, and contextual appropriateness in preventive health counseling tasks.
- **Web App Deployment:** Designed and developed an AI-driven web app with a fine-tuned OpenAI model and Streamlit, mainly using Python; Created authentication through Firebase.

- **Feature Implementation:** Built a real-time conversational interface with chat history storage and export functionality to enhance user engagement and usability.

Improving the Decision Boundary of Non-Parametric Outlier Synthesis Model

01/2025-03/2025

Research Investigator (Supervisor: Kaizhu Huang)

Kunshan, Jiangsu, China

- **Background:** Conducted enhancements to the kNN-based decision boundary of Non-Parametric Outlier Synthesis (NPOS) model to improve out-of-distribution (OOD) detection performance under high-dimensional settings.
- **Synthesis Methods:** Evaluated alternative outlier synthesis methods through GPU, including Gaussian Mixture Models, Kernel Density Estimation, and uniform sampling, via statistical metrics (FPR95, AUROC, AUPR).
- **Data Visualization:** Visualized high-dimensional embeddings and synthetic outliers using t-SNE and systematically compared k-NN distance behavior under different noise distributions and dimensionalities.
- **Limitation Analysis:** Identified limitations of Euclidean distance in high-dimensional spaces and proposed future directions, including metric learning and physics-inspired sampling.

Systematic Review: Key Factors in AI Algorithms Applied in Cancer Treatment

06/2023-03/2024

Research Assistant (Supervisor: Meifang Chen)

Kunshan, Jiangsu, China & Los Angeles, CA, U.S.

- **Background:** Investigated methodological and clinical factors incorporated in AI-driven cancer treatment prediction models to evaluate their role in advancing precision oncology and reducing care disparities.
- **Systematic Review:** Led a systematic review of 3,325+ articles, developing inclusion/exclusion criteria and structured data extraction protocols to identify key algorithmic components (e.g., LASSO, CNNs, ensemble models) and clinical covariates used in treatment prediction.
- **Comparative Algorithm Analysis:** Screened and synthesized findings via ROB2 and Cochrane to subsequently assess model types, input features (genomic, imaging, clinical).
- **Co-author collaboration:** Collaborated closely with a faculty advisor and a team member to design the review methodology, analyze findings, and co-author the manuscript, ensuring rigorous and impactful research outcomes.
- **Results:** Presented findings at the CUGH 2024 Conference (Los Angeles) via poster presentation, highlighting AI's potential to reduce cancer care disparities.

DATA SCIENTIST EXPERIENCE

Merit Interactive Co., Ltd.

06/2024-08/2024

Data Scientist Intern

Hangzhou, Zhejiang, China

- **Background:** Conducted large-scale healthcare data governance, focusing on data quality management, data integration and sharing, and data-driven decision making, etc., to improve consistency and accessibility.
- **Data Processing:** Extracted and standardized healthcare and pharmacy datasets with 100M+ data using FlinkSQL and Python through the company's own Cloud OS and Docker across multiple dimensions.
- **Data Visualization & Analysis:** Developed interactive multidimensional visualizations (e.g., medication sales, hospital operations, etc.), such as heatmaps, bar charts, and pie charts, supporting data-driven strategic marketing decisions.
- **API Creation & Testing:** Built and tested 3 customer-facing APIs for pharmaceutical dataset access; Optimized their performance and ensured reliability with Python and JavaScript; Deployed improved code to GitLab for version control.
- **Collaboration:** Collaborated cross-functionally with colleagues from non-tech backgrounds, translating analytical findings into clear, actionable insights.

TECHNICAL SKILLS

- **Programming Languages:** Python (TensorFlow, PyTorch, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Streamlit, LangChain), R (ggplot2, tidyverse, bootstrap), SQL, Slurm, Java, JavaScript, HTML, CSS
- **Machine Learning & AI:** Deep Learning (Neural Networks), Transformers, LLMs, RAG, LangChain, Hugging Face, Prompt Engineering, and Hyperparameter Tuning
- **Development Tools:** Git, VSCode, Android Studio, R Studio, Jupyter Notebook, Anaconda, Yale High-performance computing (HPC) Cluster, Cloud Computing Platforms (AWS, Azure), Jenkins, Nextflow
- **Data Visualization:** Tableau, Power BI, Canva